

Md. Gulam Mustafa

Lecturer (Physics)

Department of Science and Humanities

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Google Scholar: <https://scholar.google.com/citations?user=InR7XhAAAAAJ&hl=en>

RESEARCH INTERESTS

Advanced Functional Materials; Multiferroic and Magnetoelectric Materials; Ferrites and Functional Oxide Ceramics; Electromagnetic Materials and Composites; Doping and Compositional Engineering; Dielectric, Magnetic, Electrical, Optical and Magnetoelectric Properties; Energy Harvesting; High-Frequency and Terahertz (THz) Materials and Devices.

PUBLICATIONS

1. **Md. Gulam Mustafa**, Salma Akter, S.C. Mazumdar, “Enhanced multiferroic properties of NZCFO–BNFSO composites: Synthesis, characterization, and performance Analysis”, *Heliyon*, Volume 10, Issue 15, e35260, August 2024.
DOI: [10.1016/j.heliyon.2024.e35260](https://doi.org/10.1016/j.heliyon.2024.e35260)
2. S.C. Mazumdar, **Md. Gulam Mustafa**, “Enhancement of Multiferroic and Functional Properties of $(1-y)\text{Bi}_{0.7}\text{La}_{0.3}\text{FeO}_3+y\text{Ni}_{0.50}\text{Cu}_{0.05}\text{Zn}_{0.45}\text{Fe}_2\text{O}_4$ Composite Ceramics”, *Ceramics International*, Volume 52, Issue 13, Part A, May 2026, Pages 21540-21558.
DOI: [10.1016/j.ceramint.2026.03.232](https://doi.org/10.1016/j.ceramint.2026.03.232)
3. Tanvir Ahmed Fahim, Md. Abdullah-Al-Mamun, Joton Kumar Dea, **Md. Gulam Mustafa**, “Machine learning-driven gain prediction of a high-performance dual-wideband decoupling-enhanced graphene MIMO antenna for THz applications”, **Diamond and Related Materials**, Volume 169, Part B, Article 114121, November 2026.
DOI: [10.1016/j.diamond.2026.114121](https://doi.org/10.1016/j.diamond.2026.114121)

ONGOING RESEARCHES

1. Compact Multiresonant Graphene-Based THz Antenna with RLC Circuit Validation and ML-Assisted Efficiency Prediction for Advanced Wireless Applications. (Under Review)
2. Optimization of Multiferroic Performance in BiFeO_3 Ceramics via Nd–Sc Co-Doping. (Under Review)
3. Super-Wideband Graphene-Based THz MIMO Antenna with Machine-Learning-Based Performance Prediction for IoT- and IoNT-Enabled 6G Smart Cities.

4. Effects of Ho and Sc co-doping on the multiferroic properties of BiFeO₃.
5. Composition-Driven Structural, Optical, Dielectric, Electrical, and Magnetic Properties of Ca–Ni Co-Doped BaTiO₃ Ceramics
6. Thermoelectric and Mechanical Trade off in Metal Oxide Reinforced Portland Composite Cement: A Comparative Study of Fe₂O₃, CuO, and Co₃O₄
7. Optimizing multiferroic interactions in La and Sm co-doped BiFeO₃: Structural, Magnetic and Dielectric Perspective via computational analysis
8. ZnO-Modified Portland Composite Cement for Building Integrated Thermoelectric Energy Harvesting: Optimization, Microstructural Evolution, and Multifunctional Performance
9. Effects of high-energy ball milling and Bi₂O₃ additive on microstructure and electromagnetic properties of low-temperature-sintered Ni–Zn ferrite
10. Fabrication of Low Cost ZnO p–n Homojunction Thin Films by Solution-Based Methods for Renewable Energy and Sensor Applications

PEER REVIEW EXPERIENCE

1. Ceramics International: July 2026 – Present

- Reviewing manuscripts related to ceramics, functional materials, multiferroic materials, and related areas of materials science.
- Providing critical evaluation of manuscript quality, scientific methodology, experimental results, analysis, and overall contribution to the field.

TEACHING EXPERIENCE

1. Lecturer (Physics): 29 October 2023 – Present

Department of Science and Humanities
Bangladesh Army International University of Science and Technology (BAIUST)
Cumilla 3501, Bangladesh

2. Lecturer (Physics): 4 January 2023 – 28 October 2023

KC Model School and College
Dhaka 1230, Bangladesh

RESEARCH PROJECT EXPERIENCE & SUPERVISION

1. **Co-investigator**, Sustainable and Renewable Energy Development Authority (SREDA) Research Project (1 July 2026 – Present)
Project Title: Fabrication of Low Cost ZnO p–n Homojunction Thin Films by Solution-Based Methods for Renewable Energy and Sensor Applications.
2. **Co-supervisor**, Department of Electrical and Electronic Engineering, Bangladesh Army International University of Science and Technology (BAIUST), Cumilla 3501, Bangladesh (Semester: Spring 2025 – Fall 2025)

Capstone Design Project (EEE 400-1 & EEE 400-2) Title: Optimizing Multiferroic Interactions in La and Sm Co-doped BiFeO₃: Structural, Electronic & Magnetic Perspective via Computational Analysis.

3. **Assistant Project Director**, University Grants Commission of Bangladesh (UGC) Research Project (1 July 2022 – 30 June 2023)

Project Title: Investigation of the structural, multiferroic and transport properties of low temperature sintered $x\text{Ni}_{0.24}\text{Zn}_{0.58}\text{Cu}_{0.18}\text{Fe}_2\text{O}_4-(1-x)\text{Bi}_{0.9}\text{Nd}_{0.1}\text{Fe}_{0.95}\text{Sc}_{0.05}\text{O}_3$ multiferroic composites.

PROFESSIONAL & ACADEMIC EXPERIENCE

1. **Faculty-in-Charge: 24 December 2025 – Present**
Physics Laboratory, Department of Science and Humanities
2. **Co-Advisor: June 2024 – Present**
BAIUST Robotics Society
3. **Member**, BAIUST Central Examination Committee (Semester: Fall 2023).

EDUCATIONAL BACKGROUND

Master of Science (M.Sc.) in Physics (Thesis) — Comilla University, Cumilla, Bangladesh
Session: 2019–2020 | CGPA: **3.77** in the scale of 4.00

Bachelor of Science (B.Sc.) in Physics — Comilla University, Cumilla, Bangladesh
Session: 2015–2016 | CGPA: **3.56** in the scale of 4.00

Higher Secondary Certificate (HSC) — Bir Shreshtha Noor Mohammad Public College, Dhaka, Bangladesh
Session: 2012–2013 | GPA: **5.00** in the scale of 5.00

Secondary School Certificate (SSC) — West End High School, Dhaka, Bangladesh
Session: 2010–2011 | GPA: **5.00** in the scale of 5.00

FELLOWSHIP & GRANT

- National Science and Technology (NST) Fellowship 2022-2023.
- University Grants Commission of Bangladesh (UGC) Research Grant 2022-2023.
- Sustainable and Renewable Energy Development Authority (SREDA) Research Grant 2025-2026.

TECHNICAL SKILLS

Microsoft Office: Expert in *Microsoft Word*, *Microsoft Excel*, and *Microsoft PowerPoint* for academic writing, data organization and analysis, presentation development, and professional documentation.

Scientific Data Analysis and Visualization: Proficient in *OriginLab* for scientific data processing, graphical representation, curve fitting, and analysis of experimental results.

Engineering Design and Simulation: Proficient in *CST Studio Suite* for electromagnetic simulation, antenna design, modeling, and performance analysis.

Technical Drawing and Diagramming: Proficient in *Microsoft Visio* for creating scientific diagrams, flowcharts, technical illustrations, and professional schematics.

Reference Management: Proficient in *Mendeley* for reference management, citation organization, and preparation of academic manuscripts.

LANGUAGE SKILLS

Have good command both in written and spoken English.

Speak fluently in Bengali without uttering as it is my mother tongue.

REFEREES

Dr. Sajal Chandra Mazumdar
Professor & Chairman, Department of Physics
Comilla University, Cumilla 3506, Bangladesh
Cell: +8801729283861
email: sajal@yahoo.com

Dr. Md. Kawchar Ahmed Patwary
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DECLARATION OF AUTHENTICITY

I do hereby declare that all information here is true to my knowledge. If required and whereapplicable, this document can be supported by appropriate authentic certificates/Paper.



(Md. Gulam Mustafa)